設定軟體
Cluster and storage switches

NetApp
March 22, 2024

This PDF was generated from https://docs.netapp.com/zh-tw/ontap-systems-switches/switch-cisco-92300/install-nxos-overview.html on March 22, 2024. Always check docs.netapp.com for the latest.

目錄

設定軟體 1
準備安裝NX-OS軟體和參考組態檔（RCF） 1
安裝NX-OS軟體 8
安裝參考組態檔（RCF） 18
乙太網路交換器健全狀況監控記錄收集 38
設定 SNMPv3 40

設定軟體

準備安裝NX-OS軟體和參考組態檔（RCF）

在安裝NX-OS軟體和參考組態檔（RCF）之前、請遵循此程序。

您需要的產品

- 完全正常運作的叢集（記錄檔中沒有錯誤或類似問題）。
- 如需適當的軟體與升級指南、請參閱 "Cisco Nexus 9000系列交換器"。

關於範例

本程序中的範例使用兩個節點。這些節點使用兩個10GbE叢集互連連接埠 `e0a` 和 `e0b`。請參閱 "Hardware Universe" 驗證平台上的叢集連接埠是否正確。

本程序中的範例使用下列交換器和節點命名法：

- 兩個Cisco交換器的名稱分別為「CS1」和「CS2」。
- 節點名稱是「node1」和「node2」。
- 叢集LIF的名稱為節點1的「node1_clus1」和「node1_clus2」、節點2的「node2_clus1」和「node2_clus2」。
- 「cluster1：*：>」提示會指出叢集的名稱。

關於這項工作

此程序需要同時使用ONTAP 支援指令和Cisco Nexus 9000系列交換器的指令；ONTAP 除非另有說明、否則會使用支援指令。命令輸出可能會因ONTAP 不同版本的不相同而有所差異。

步驟

1. 將權限等級變更為進階、並在系統提示您繼續時輸入* y*：

   "進階權限"

   出現進階提示（「*>」）。

2. 如果AutoSupport 此叢集啟用了「支援」功能、請叫用AutoSupport 下列訊息來禁止自動建立案例：

   「系統節點AutoSupport 不完整地叫用節點*-type all -Message MAn=xh」

   其中_x_是維護時段的持續時間（以小時為單位）。

   > 此資訊可通知技術支援人員執行此維護工作、以便在維護期間抑制自動建立案例。AutoSupport

   下列命令會禁止自動建立兩小時的個案：

```
cluster1:> **system node autosupport invoke -node * -type all -message
MAINT=2h**
```

3. 顯示每個叢集互連交換器的每個節點已設定多少個叢集互連介面：「network device-dDiscovery show -protocol cup」

   顯示範例

```
cluster1::*> network device-discovery show -protocol cdp

 Node/      Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface
Platform
----------- ------ ------------------------- ----------------
----------------
node2      /cdp
            e0a    cs1                       Eth1/2            N9K-
C92300YC
            e0b    cs2                       Eth1/2            N9K-
C92300YC
node1      /cdp
            e0a    cs1                       Eth1/1            N9K-
C92300YC
            e0b    cs2                       Eth1/1            N9K-
C92300YC

4 entries were displayed.
```

4. 檢查每個叢集介面的管理或作業狀態。

   a. 顯示網路連接埠屬性：`network port show -ipspace Cluster`

顯示範例

```
cluster1::*> network port show -ipspace Cluster

Node: node2
                                                      Speed(Mbps)
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status
--------- ------------ ---------------- ---- ---- -----------
--------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy
e0b       Cluster      Cluster          up   9000  auto/10000
healthy

Node: node1
                                                      Speed(Mbps)
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status
--------- ------------ ---------------- ---- ---- -----------
--------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy
e0b       Cluster      Cluster          up   9000  auto/10000
healthy

4 entries were displayed.
```

b. 顯示有關生命的資訊：「網路介面show -vserver叢集」

顯示範例

```
cluster1::*> network interface show -vserver Cluster

            Logical    Status     Network            Current
Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node
Port    Home
----------- ---------- ---------- ------------------
------------- ------- ----
Cluster
            node1_clus1  up/up    169.254.209.69/16  node1
e0a     true
            node1_clus2  up/up    169.254.49.125/16  node1
e0b     true
            node2_clus1  up/up    169.254.47.194/16  node2
e0a     true
            node2_clus2  up/up    169.254.19.183/16  node2
e0b     true

4 entries were displayed.
```

5. Ping遠端叢集LIF：

   「叢集ping叢集節點節點名稱」

顯示範例

```
cluster1::*> cluster ping-cluster -node node2
Host is node2
Getting addresses from network interface table...
Cluster node1_clus1 169.254.209.69 node1     e0a
Cluster node1_clus2 169.254.49.125 node1     e0b
Cluster node2_clus1 169.254.47.194 node2     e0a
Cluster node2_clus2 169.254.19.183 node2     e0b
Local = 169.254.47.194 169.254.19.183
Remote = 169.254.209.69 169.254.49.125
Cluster Vserver Id = 4294967293
Ping status:

Basic connectivity succeeds on 4 path(s)
Basic connectivity fails on 0 path(s)

Detected 9000 byte MTU on 4 path(s):
    Local 169.254.19.183 to Remote 169.254.209.69
    Local 169.254.19.183 to Remote 169.254.49.125
    Local 169.254.47.194 to Remote 169.254.209.69
    Local 169.254.47.194 to Remote 169.254.49.125
Larger than PMTU communication succeeds on 4 path(s)
RPC status:
2 paths up, 0 paths down (tcp check)
2 paths up, 0 paths down (udp check)
```

6. 驗證是否已在所有叢集生命體上啟用自動還原命令：

   「網路介面show -vserver叢集-功能 變數自動回復」

顯示範例

```
cluster1::*> network interface show -vserver Cluster -fields auto-
revert

          Logical
Vserver   Interface     Auto-revert
--------- ------------- ------------
Cluster
          node1_clus1   true
          node1_clus2   true
          node2_clus1   true
          node2_clus2   true

4 entries were displayed.
```

7. 對於支援支援支援支援更新版本的版本、請啟用叢集交換器健全狀況監控記錄收集功能、以便使用下列命令來收集交換器相關的記錄檔：ONTAP

   「系統叢集交換器記錄設定密碼」和「系統叢集交換器記錄啟用收集」

顯示範例

```
cluster1::*> system cluster-switch log setup-password
Enter the switch name: <return>
The switch name entered is not recognized.
Choose from the following list:
cs1
cs2

cluster1::*> system cluster-switch log setup-password

Enter the switch name: cs1
RSA key fingerprint is
e5:8b:c6:dc:e2:18:18:09:36:63:d9:63:dd:03:d9:cc
Do you want to continue? {y|n}::[n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> system cluster-switch log setup-password

Enter the switch name: cs2
RSA key fingerprint is
57:49:86:a1:b9:80:6a:61:9a:86:8e:3c:e3:b7:1f:b1
Do you want to continue? {y|n}:: [n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> system cluster-switch log enable-collection

Do you want to enable cluster log collection for all nodes in the
cluster?
{y|n}: [n] y

Enabling cluster switch log collection.

cluster1::*>
```

如果這些命令中有任何一個出現錯誤、請聯絡NetApp支援部門。

接下來呢？

"安裝NX-OS軟體"。

安裝NX-OS軟體

請遵循此程序、在Nexus 2300YC交換器上安裝NX-OS軟體。

NX-OS是Cisco Systems提供的Nexus系列乙太網路交換器和MDS系列光纖通道（FC）儲存區域網路交換器的網路作業系統。

檢閱要求

支援的連接埠和節點連線

- Nexus 2300YC交換器支援的交換器間連結（ISL）為連接埠1/65和1/66。
- Nexus 2300YC交換器支援的節點連線為連接埠1/1到1/66。

您需要的產品

- 適用於您交換器NetApp 支援網站 的NetApp Cisco NX-OS軟體、可從下列網站取得："mysupport.netapp.com"
- 完全正常運作的叢集（記錄檔中沒有錯誤或類似問題）。
- "Cisco乙太網路交換器頁面"。請參閱交換器相容性表、以瞭解支援ONTAP 的功能表和NX-OS版本。

安裝軟體

本程序中的範例使用兩個節點、但叢集中最多可有24個節點。

關於範例

本程序中的範例使用下列交換器和節點命名法：

- Nexus 92300YC交換器名稱為「CS1」和「CS2」。
- 此程序中使用的範例會在第二個交換器上開始升級、即：* *CS2**。
- 叢集LIF名稱為節點1的「node1_clus1」和「node1_clus2」、節點2的「node2_clus1」和「node2_clus2」。
- IPspace名稱為「叢集」。
- 「cluster1：*：>」提示會指出叢集的名稱。
- 每個節點上的叢集連接埠名稱為「e0a」和「e0b」。

  請參閱 "*SUR1_* Hardware Universe" 以取得平台所支援的實際叢集連接埠。

步驟

1. 將叢集交換器連接至管理網路。
2. 使用「ping」命令來驗證與裝載NX-OS軟體和RCF之伺服器的連線。

顯示範例

此範例可驗證交換器是否能以IP位址172.19.2.1連至伺服器：

```
cs2# ping 172.19.2.1
Pinging 172.19.2.1 with 0 bytes of data:

Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
```

3. 將NX-OS軟體和EPLD映像複製到Nexus 2300YC交換器。

顯示範例

```
cs2# copy sftp: bootflash: vrf management
Enter source filename: /code/nxos.9.2.2.bin
Enter hostname for the sftp server: 172.19.2.1
Enter username: user1

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/nxos.9.2.2.bin  /bootflash/nxos.9.2.2.bin
/code/nxos.9.2.2.bin  100% 1261MB   9.3MB/s   02:15
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.


cs2# copy sftp: bootflash: vrf management
Enter source filename: /code/n9000-epld.9.2.2.img
Enter hostname for the sftp server: 172.19.2.1
Enter username: user1

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/n9000-epld.9.2.2.img  /bootflash/n9000-
epld.9.2.2.img
/code/n9000-epld.9.2.2.img  100%  161MB   9.5MB/s   00:16
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
```

4. 驗證NX-OS軟體的執行版本：

   《如何版本》

顯示範例

```
cs2# show version
Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
Copyright (C) 2002-2018, Cisco and/or its affiliates.
All rights reserved.
The copyrights to certain works contained in this software are
owned by other third parties and used and distributed under their
own
licenses, such as open source.  This software is provided "as is,"
and unless
otherwise stated, there is no warranty, express or implied,
including but not
limited to warranties of merchantability and fitness for a
particular purpose.
Certain components of this software are licensed under
the GNU General Public License (GPL) version 2.0 or
GNU General Public License (GPL) version 3.0  or the GNU
Lesser General Public License (LGPL) Version 2.1 or
Lesser General Public License (LGPL) Version 2.0.
A copy of each such license is available at
http://www.opensource.org/licenses/gpl-2.0.php and
http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.

Software
  BIOS: version 05.31
  NXOS: version 9.2(1)
  BIOS compile time:  05/17/2018
  NXOS image file is: bootflash:///nxos.9.2.1.bin
  NXOS compile time:  7/17/2018 16:00:00 [07/18/2018 00:21:19]


Hardware
  cisco Nexus9000 C92300YC Chassis
  Intel(R) Xeon(R) CPU D-1526 @ 1.80GHz with 16337884 kB of memory.
  Processor Board ID FDO220329V5

  Device name: cs2
  bootflash:  115805356 kB
Kernel uptime is 0 day(s), 4 hour(s), 23 minute(s), 11 second(s)

Last reset at 271444 usecs after Wed Apr 10 00:25:32 2019
  Reason: Reset Requested by CLI command reload
```

```
  System version: 9.2(1)
  Service:

plugin
  Core Plugin, Ethernet Plugin

Active Package(s):

cs2#
```

5. 安裝NX-OS映像。

   安裝映像檔會在每次重新開機時載入映像檔。

顯示範例

```
cs2# install all nxos bootflash:nxos.9.2.2.bin

Installer will perform compatibility check first. Please wait.
Installer is forced disruptive

Verifying image bootflash:/nxos.9.2.2.bin for boot variable "nxos".
[] 100% -- SUCCESS

Verifying image type.
[] 100% -- SUCCESS

Preparing "nxos" version info using image bootflash:/nxos.9.2.2.bin.
[] 100% -- SUCCESS

Preparing "bios" version info using image bootflash:/nxos.9.2.2.bin.
[] 100% -- SUCCESS

Performing module support checks.
[] 100% -- SUCCESS

Notifying services about system upgrade.
[] 100% -- SUCCESS



Compatibility check is done:
Module  bootable       Impact     Install-type  Reason
------  --------  --------------- ------------  ------
  1       yes      disruptive         reset     default upgrade is
not hitless



Images will be upgraded according to following table:

Module   Image         Running-Version(pri:alt            New-
Version         Upg-Required
------  --------  ---------------------------------------
------------------  ------------
  1       nxos                                  9.2(1)
9.2(2)         yes
  1       bios    v05.31(05/17/2018):v05.28(01/18/2018)
v05.33(09/08/2018)       yes
```

```
Switch will be reloaded for disruptive upgrade.
Do you want to continue with the installation (y/n)?  [n] y


Install is in progress, please wait.

Performing runtime checks.
[] 100% -- SUCCESS

Setting boot variables.
[] 100% -- SUCCESS

Performing configuration copy.
[] 100% -- SUCCESS

Module 1: Refreshing compact flash and upgrading
bios/loader/bootrom.
Warning: please do not remove or power off the module at this time.
[] 100% -- SUCCESS
2019 Apr 10 04:59:35 cs2 %$ VDC-1 %$ %VMAN-2-ACTIVATION_STATE:
Successfully deactivated virtual service 'guestshell+'

Finishing the upgrade, switch will reboot in 10 seconds.
```

6. 在交換器重新開機後、驗證新版本的NX-OS軟體：

《如何版本》

顯示範例

```
cs2# show version

Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
Copyright (C) 2002-2018, Cisco and/or its affiliates.
All rights reserved.
The copyrights to certain works contained in this software are
owned by other third parties and used and distributed under their
own
licenses, such as open source.  This software is provided "as is,"
and unless
otherwise stated, there is no warranty, express or implied,
including but not
limited to warranties of merchantability and fitness for a
particular purpose.
Certain components of this software are licensed under
the GNU General Public License (GPL) version 2.0 or
GNU General Public License (GPL) version 3.0  or the GNU
Lesser General Public License (LGPL) Version 2.1 or
Lesser General Public License (LGPL) Version 2.0.
A copy of each such license is available at
http://www.opensource.org/licenses/gpl-2.0.php and
http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.

Software
  BIOS: version 05.33
  NXOS: version 9.2(2)
  BIOS compile time:  09/08/2018
  NXOS image file is: bootflash:///nxos.9.2.2.bin
  NXOS compile time:  11/4/2018 21:00:00 [11/05/2018 06:11:06]


Hardware
  cisco Nexus9000 C92300YC Chassis
  Intel(R) Xeon(R) CPU D-1526 @ 1.80GHz with 16337884 kB of memory.
  Processor Board ID FDO220329V5

  Device name: cs2
  bootflash:  115805356 kB
  Kernel uptime is 0 day(s), 0 hour(s), 3 minute(s), 52 second(s)

Last reset at 182004 usecs after Wed Apr 10 04:59:48 2019
```

```
  Reason: Reset due to upgrade
  System version: 9.2(1)
  Service:

plugin
  Core Plugin, Ethernet Plugin

Active Package(s):
```

7. 升級EPLD映像、然後重新啟動交換器。

顯示範例

```
cs2# show version module 1 epld

EPLD Device                      Version
---------------------------------------
MI FPGA                           0x7
IO FPGA                           0x17
MI FPGA2                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2

cs2# install epld bootflash:n9000-epld.9.2.2.img module 1
Compatibility check:
Module        Type         Upgradable        Impact   Reason
------  -----------------  ----------    ----------   ------
     1            SUP           Yes       disruptive  Module
Upgradable

Retrieving EPLD versions.... Please wait.
Images will be upgraded according to following table:
Module  Type   EPLD              Running-Version   New-Version  Upg-
Required
------  ----  -------------      ---------------   -----------
------------
     1   SUP  MI FPGA                   0x07        0x07
No
     1   SUP  IO FPGA                   0x17        0x19
Yes
     1   SUP  MI FPGA2                  0x02        0x02
No
The above modules require upgrade.
The switch will be reloaded at the end of the upgrade
Do you want to continue (y/n) ?  [n] y

Proceeding to upgrade Modules.

Starting Module 1 EPLD Upgrade

Module 1 : IO FPGA [Programming] : 100.00% (     64 of      64
sectors)
Module 1 EPLD upgrade is successful.
Module        Type  Upgrade-Result
------  ------------------  --------------
```

```
      1          SUP          Success


EPLDs upgraded.

Module 1 EPLD upgrade is successful.
```

8. 交換器重新開機後、再次登入、並確認新版的EPLD已成功載入。

顯示範例

```
cs2# *show version module 1 epld*

EPLD Device                        Version
---------------------------------------
MI FPGA                             0x7
IO FPGA                             0x19
MI FPGA2                            0x2
GEM FPGA                            0x2
GEM FPGA                            0x2
GEM FPGA                            0x2
GEM FPGA                            0x2
```

接下來呢？

"安裝參考組態檔案"

安裝參考組態檔（RCF）

您可以在初次設定Nexus 2300YC交換器之後安裝RCF。您也可以使用此程序來升級RCF版本。

關於這項工作

本程序中的範例使用下列交換器和節點命名法：

- 兩個Cisco交換器的名稱分別為「CS1」和「CS2」。
- 節點名稱是「node1」和「node2」。
- 叢集LIF名稱為 `node1_clus1`、`node1_clus2`、`node2_clus1`和 `node2_clus2`。
- 「cluster1：*：>」提示會指出叢集的名稱。

- 此程序需要同時使用ONTAP 支援指令和 "Cisco Nexus 9000系列交換器"；ONTAP 除非另有說明、否則會使用指令。
- 執行此程序之前、請先確定您目前已備份交換器組態。
- 在此程序期間、不需要運作中的交換器間連結（ISL）。這是因為RCF版本變更可能會暫時影響ISL連線。為確保叢集作業不中斷、下列程序會在目標交換器上執行步驟時、將所有叢集生命體移轉至作業夥伴交換器。

步驟

1. 顯示連接至叢集交換器的每個節點上的叢集連接埠：「network device-dDiscovery show」

   顯示範例

```
cluster1::*> *network device-discovery show*
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface
Platform
----------- ------ ------------------------- ----------------
------------
node1/cdp
            e0a    cs1                       Ethernet1/1/1     N9K-
C92300YC
            e0b    cs2                       Ethernet1/1/1     N9K-
C92300YC
node2/cdp
            e0a    cs1                       Ethernet1/1/2     N9K-
C92300YC
            e0b    cs2                       Ethernet1/1/2     N9K-
C92300YC
cluster1::*>
```

2. 檢查每個叢集連接埠的管理和作業狀態。
   a. 確認所有叢集連接埠都正常運作：`network port show -ipspace Cluster`

顯示範例

```
cluster1::*> *network port show -ipspace Cluster*

Node: node1

Ignore
                                                      Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0c       Cluster      Cluster          up   9000  auto/100000
healthy false
e0d       Cluster      Cluster          up   9000  auto/100000
healthy false

Node: node2

Ignore
                                                      Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0c       Cluster      Cluster          up   9000  auto/100000
healthy false
e0d       Cluster      Cluster          up   9000  auto/100000
healthy false
cluster1::*>
```

b. 確認所有叢集介面（l生命）都位於主連接埠： `network interface show -vserver Cluster`

顯示範例

```
cluster1::*> *network interface show -vserver Cluster*
            Logical            Status     Network
Current      Current Is
Vserver     Interface          Admin/Oper Address/Mask      Node
Port    Home
----------- ------------------ ---------- -----------------
------------ ------- ----
Cluster
            node1_clus1        up/up      169.254.3.4/23    node1
e0c     true
            node1_clus2        up/up      169.254.3.5/23    node1
e0d     true
            node2_clus1        up/up      169.254.3.8/23    node2
e0c     true
            node2_clus2        up/up      169.254.3.9/23    node2
e0d     true
cluster1::*>
```

c. 驗證叢集是否顯示兩個叢集交換器的資訊：「系統叢集交換器show -is監控、啟用作業的true」

顯示範例

```
cluster1::*> *system cluster-switch show -is-monitoring-enabled
-operational true*
Switch                      Type               Address
Model
--------------------------- ------------------ ----------------
---------------
cs1                         cluster-network    10.233.205.92
N9K-C92300YC
     Serial Number: FOXXXXXXXGS
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                    9.3(4)
    Version Source: CDP

cs2                         cluster-network    10.233.205.93
N9K-C92300YC
     Serial Number: FOXXXXXXXGD
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                    9.3(4)
    Version Source: CDP

2 entries were displayed.
```

3. 停用叢集生命體上的自動還原。

```
cluster1::*> network interface modify -vserver Cluster -lif * -auto
-revert false
```

4. 在叢集交換器CS2上、關閉連接至節點叢集連接埠的連接埠。

```
cs2(config)# interface e1/1-64
cs2(config-if-range)# shutdown
```

5. 確認叢集連接埠已移轉至叢集交換器CS1上裝載的連接埠。這可能需要幾秒鐘的時間。 `network interface show -vserver Cluster`

顯示範例

```
cluster1::*> *network interface show -vserver Cluster*
            Logical           Status     Network            Current
Current Is
Vserver     Interface         Admin/Oper Address/Mask       Node
Port    Home
----------- ----------------- ---------- ------------------
------------- ------- ----
Cluster
            node1_clus1       up/up      169.254.3.4/23     node1
e0c     true
            node1_clus2       up/up      169.254.3.5/23     node1
e0c     false
            node2_clus1       up/up      169.254.3.8/23     node2
e0c     true
            node2_clus2       up/up      169.254.3.9/23     node2
e0c     false
cluster1::*>
```

6. 驗證叢集是否正常：「叢集顯示」

顯示範例

```
cluster1::*> *cluster show*
Node           Health  Eligibility   Epsilon
-------------- ------- ------------  -------
node1          true    true          false
node2          true    true          false
cluster1::*>
```

7. 如果您尚未這麼做、請將下列命令的輸出複製到文字檔、以儲存目前交換器組態的複本：

「如何執行設定」

8. 清除交換器CS2上的組態、然後執行基本設定。

> 更新或套用新的RCF時、您必須清除交換器設定並執行基本組態。您必須連線至交換器序列主控台連接埠、才能再次設定交換器。

a. 清理組態：

顯示範例

```
(cs2)# write erase

Warning: This command will erase the startup-configuration.

Do you wish to proceed anyway? (y/n)  [n]  y
```

b. 重新啟動交換器：

顯示範例

```
(cs2)# reload

Are you sure you would like to reset the system? (y/n) y
```

9. 使用下列傳輸傳輸協定之一、將RCF複製到交換器CS2的bootflash：FTP、TFTP、SFTP或scp。如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列交換器" 指南：

   本範例顯示使用TFTP將RCF複製到交換器CS2上的bootFlash：

```
cs2# copy tftp: bootflash: vrf management
Enter source filename: /code/Nexus_92300YC_RCF_v1.0.2.txt
Enter hostname for the tftp server: 172.19.2.1
Enter username: user1

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
tftp> progress
Progress meter enabled
tftp> get /code/Nexus_92300YC_RCF_v1.0.2.txt /bootflash/nxos.9.2.2.bin
/code/Nexus_92300YC_R  100% 9687   530.2KB/s   00:00
tftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
```

10. 將先前下載的RCF套用至bootFlash。

    如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列交換器" 指南：

    此範例顯示RCF檔案 `Nexus_92300YC_RCF_v1.0.2.txt` 安裝在交換器CS2上：

```
cs2# copy Nexus_92300YC_RCF_v1.0.2.txt running-config echo-commands

Disabling ssh: as its enabled right now:
 generating ecdsa key(521 bits)......
generated ecdsa key

Enabling ssh: as it has been disabled
 this command enables edge port type (portfast) by default on all
interfaces. You
 should now disable edge port type (portfast) explicitly on switched
ports leading to hubs,
 switches and bridges as they may create temporary bridging loops.

Edge port type (portfast) should only be enabled on ports connected to a
single
 host. Connecting hubs, concentrators, switches, bridges, etc...  to
this
 interface when edge port type (portfast) is enabled, can cause
temporary bridging loops.
 Use with CAUTION

Edge Port Type (Portfast) has been configured on Ethernet1/1 but will
only
 have effect when the interface is in a non-trunking mode.

...

Copy complete, now saving to disk (please wait)...
Copy complete.
```

11. 在交換器上確認RCF已成功合併：

    「如何執行設定」

```
cs2# show running-config
!Command: show running-config
!Running configuration last done at: Wed Apr 10 06:32:27 2019
!Time: Wed Apr 10 06:36:00 2019

version 9.2(2) Bios:version 05.33
switchname cs2
vdc cs2 id 1
  limit-resource vlan minimum 16 maximum 4094
  limit-resource vrf minimum 2 maximum 4096
  limit-resource port-channel minimum 0 maximum 511
  limit-resource u4route-mem minimum 248 maximum 248
  limit-resource u6route-mem minimum 96 maximum 96
  limit-resource m4route-mem minimum 58 maximum 58
  limit-resource m6route-mem minimum 8 maximum 8

feature lacp

no password strength-check
username admin password 5
$5$HY9Kk3F9$YdCZ8iQJ1RtoiEFa0sKP5IO/LNG1k9C4lSJfi5kesl
6  role network-admin
ssh key ecdsa 521

banner motd #

*
*
*  Nexus 92300YC Reference Configuration File (RCF) v1.0.2 (10-19-2018)
*
*
*
*  Ports 1/1  - 1/48: 10GbE Intra-Cluster Node Ports
*
*  Ports 1/49 - 1/64: 40/100GbE Intra-Cluster Node Ports
*
*  Ports 1/65 - 1/66: 40/100GbE Intra-Cluster ISL Ports
*
*
*
```

第一次套用RCF時、預期會出現*錯誤：無法寫入VSH命令*訊息、因此可以忽略。

1. [[step12]確認RCF檔案為正確的更新版本： `show running-config`

當您檢查輸出以確認您擁有正確的RCF時、請確定下列資訊正確無誤：

- RCF橫幅
- 節點和連接埠設定
- 自訂

  輸出會因站台組態而異。請檢查連接埠設定、並參閱版本說明、以瞭解您安裝的RCF的任何特定變更。

2. 驗證RCF版本和交換器設定是否正確之後、請將執行組態檔複製到啟動組態檔。

   如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列交換器" 指南：

```
cs2# copy running-config startup-config
[] 100% Copy complete
```

3. 重新開機交換器CS2。您可以在交換器重新開機時忽略節點上報告的「叢集連接埠當機」事件。

```
cs2# reload
This command will reboot the system. (y/n)?  [n] y
```

4. 驗證叢集上叢集連接埠的健全狀況。

   a. 驗證叢集中所有節點的e0d連接埠是否正常運作： `network port show -ipspace Cluster`

顯示範例

```
cluster1::*> *network port show -ipspace Cluster*

Node: node1

Ignore
                                                      Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy  false
e0b       Cluster      Cluster          up   9000  auto/10000
healthy  false

Node: node2

Ignore
                                                      Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy  false
e0b       Cluster      Cluster          up   9000  auto/10000
healthy  false
```

b. 驗證叢集的交換器健全狀況（這可能不會顯示交換器CS2、因為LIF不是位於e0d上）。

顯示範例

```
cluster1::*> *network device-discovery show -protocol cdp*
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface
Platform
----------- ------ ------------------------- -----------------
------------
node1/cdp
            e0a    cs1                       Ethernet1/1
N9K-C92300YC
            e0b    cs2                       Ethernet1/1
N9K-C92300YC
node2/cdp
            e0a    cs1                       Ethernet1/2
N9K-C92300YC
            e0b    cs2                       Ethernet1/2
N9K-C92300YC

cluster1::*> *system cluster-switch show -is-monitoring-enabled
-operational true*
Switch                      Type               Address
Model
--------------------------- ------------------ ----------------
------------
cs1                         cluster-network    10.233.205.90
N9K-C92300YC
     Serial Number: FOXXXXXXXGD
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                    9.3(4)
    Version Source: CDP

cs2                         cluster-network    10.233.205.91
N9K-C92300YC
     Serial Number: FOXXXXXXXGS
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                    9.3(4)
    Version Source: CDP

2 entries were displayed.
```

您可能會在CS1交換器主控台觀察下列輸出、視先前載入交換器的RCF版本而定

```
2020 Nov 17 16:07:18 cs1 %$ VDC-1 %$ %STP-2-
UNBLOCK_CONSIST_PORT: Unblocking port port-channel1 on
VLAN0092. Port consistency restored.
2020 Nov 17 16:07:23 cs1 %$ VDC-1 %$ %STP-2-BLOCK_PVID_PEER:
Blocking port-channel1 on VLAN0001. Inconsistent peer vlan.
2020 Nov 17 16:07:23 cs1 %$ VDC-1 %$ %STP-2-BLOCK_PVID_LOCAL:
Blocking port-channel1 on VLAN0092. Inconsistent local vlan.
```

5. 在叢集交換器CS1上、關閉連接至節點叢集連接埠的連接埠。

   下列範例使用步驟1的介面輸出範例：

```
cs1(config)# interface e1/1-64
cs1(config-if-range)# shutdown
```

6. 驗證叢集LIF是否已移轉至交換器CS2上裝載的連接埠。這可能需要幾秒鐘的時間。 `network interface show -vserver Cluster`

   顯示範例

```
cluster1::*> *network interface show -vserver Cluster*
            Logical          Status     Network            Current
Current Is
Vserver     Interface        Admin/Oper Address/Mask       Node
Port    Home
----------- ---------------- ---------- ------------------
----------------- ------- ----
Cluster
            node1_clus1      up/up      169.254.3.4/23     node1
e0d     false
            node1_clus2      up/up      169.254.3.5/23     node1
e0d     true
            node2_clus1      up/up      169.254.3.8/23     node2
e0d     false
            node2_clus2      up/up      169.254.3.9/23     node2
e0d     true
cluster1::*>
```

7. 驗證叢集是否正常：「叢集顯示」

顯示範例

```
cluster1::*> *cluster show*
Node           Health   Eligibility   Epsilon
-------------- -------- ------------- -------
node1          true     true          false
node2          true     true          false
cluster1::*>
```

8. 在交換器CS1上重複步驟7至14。
9. 在叢集生命體上啟用自動還原。

```
cluster1::*> network interface modify -vserver Cluster -lif * -auto
-revert True
```

10. 重新開機交換器CS1。您可以這樣做、觸發叢集生命期以恢復到其主連接埠。您可以在交換器重新開機時忽略節點上報告的「叢集連接埠當機」事件。

```
cs1# reload
This command will reboot the system. (y/n)?  [n] y
```

11. 驗證連接至叢集連接埠的交換器連接埠是否正常運作。

```
cs1# show interface brief | grep up
.
.
Ethernet1/1      1       eth  access up      none
10G(D) --
Ethernet1/2      1       eth  access up      none
10G(D) --
Ethernet1/3      1       eth  trunk  up      none
100G(D) --
Ethernet1/4      1       eth  trunk  up      none
100G(D) --
.
.
```

12. 驗證CS1和CS2之間的ISL是否正常運作：「How port-channel Summary（顯示連接埠通道摘要）」

顯示範例

```
cs1# *show port-channel summary*
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
----------------------------------------------------------------------
------------
Group Port-       Type     Protocol  Member Ports
      Channel
----------------------------------------------------------------------
------------
1     Po1(SU)     Eth      LACP      Eth1/65(P)   Eth1/66(P)
cs1#
```

13. 驗證叢集生命區是否已還原至其主連接埠： `network interface show -vserver Cluster`

顯示範例

```
cluster1::*> *network interface show -vserver Cluster*
            Logical       Status     Network            Current
Current Is
Vserver     Interface     Admin/Oper Address/Mask       Node
Port    Home
----------- ------------- ---------- ------------------
------------- ------- ----
Cluster
            node1_clus1   up/up      169.254.3.4/23     node1
e0d     true
            node1_clus2   up/up      169.254.3.5/23     node1
e0d     true
            node2_clus1   up/up      169.254.3.8/23     node2
e0d     true
            node2_clus2   up/up      169.254.3.9/23     node2
e0d     true
cluster1::*>
```

14. 驗證叢集是否正常：「叢集顯示」

顯示範例

```
cluster1::*> *cluster show*
Node           Health  Eligibility   Epsilon
-------------- ------- ------------- -------
node1          true    true          false
node2          true    true          false
```

15. Ping遠端叢集介面以驗證連線能力：「叢集ping叢集節點local」

顯示範例

```
cluster1::*> *cluster ping-cluster -node local*
Host is node1
Getting addresses from network interface table...
Cluster node1_clus1 169.254.3.4 node1 e0a
Cluster node1_clus2 169.254.3.5 node1 e0b
Cluster node2_clus1 169.254.3.8 node2 e0a
Cluster node2_clus2 169.254.3.9 node2 e0b
Local = 169.254.1.3 169.254.1.1
Remote = 169.254.1.6 169.254.1.7 169.254.3.4 169.254.3.5 169.254.3.8
169.254.3.9
Cluster Vserver Id = 4294967293
Ping status:
............
Basic connectivity succeeds on 12 path(s)
Basic connectivity fails on 0 path(s)
................................................
Detected 9000 byte MTU on 12 path(s):
    Local 169.254.1.3 to Remote 169.254.1.6
    Local 169.254.1.3 to Remote 169.254.1.7
    Local 169.254.1.3 to Remote 169.254.3.4
    Local 169.254.1.3 to Remote 169.254.3.5
    Local 169.254.1.3 to Remote 169.254.3.8
    Local 169.254.1.3 to Remote 169.254.3.9
    Local 169.254.1.1 to Remote 169.254.1.6
    Local 169.254.1.1 to Remote 169.254.1.7
    Local 169.254.1.1 to Remote 169.254.3.4
    Local 169.254.1.1 to Remote 169.254.3.5
    Local 169.254.1.1 to Remote 169.254.3.8
    Local 169.254.1.1 to Remote 169.254.3.9
Larger than PMTU communication succeeds on 12 path(s)
RPC status:
6 paths up, 0 paths down (tcp check)
6 paths up, 0 paths down (udp check)
```

適用於更新版本的更新版本**ONTAP**

對於更新的版本、請使用下列命令啟用叢集交換器健全狀況監視器記錄收集功能、以收集交換器相關的記錄檔：ONTAP `system switch ethernet log setup-password` 和 `system switch ethernet log enable-collection`

輸入： `system switch ethernet log setup-password`

```
cluster1::*> system switch ethernet log setup-password
Enter the switch name: <return>
The switch name entered is not recognized.
Choose from the following list:
cs1
cs2

cluster1::*> system switch ethernet log setup-password

Enter the switch name: cs1
RSA key fingerprint is e5:8b:c6:dc:e2:18:18:09:36:63:d9:63:dd:03:d9:cc
Do you want to continue? {y|n}::[n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> system switch ethernet log setup-password
Enter the switch name: cs2
RSA key fingerprint is 57:49:86:a1:b9:80:6a:61:9a:86:8e:3c:e3:b7:1f:b1
Do you want to continue? {y|n}:: [n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>
```

接著： `system switch ethernet log enable-collection`

```
cluster1::*> system switch ethernet log enable-collection

Do you want to enable cluster log collection for all nodes in the
cluster?
{y|n}: [n] y

Enabling cluster switch log collection.

cluster1::*>
```

適用於更新版本的更新版本**ONTAP**

對於支援支援支援支援更新版本的版本、請啟用叢集交換器健全狀況監控記錄收集功能、以便使用下列命令來收集交換器相關的記錄檔：ONTAP

「系統叢集交換器記錄設定密碼」和「系統叢集交換器記錄啟用收集」

輸入：「ystem叢集交換器記錄設定密碼」

```
cluster1::*> system cluster-switch log setup-password
Enter the switch name: <return>
The switch name entered is not recognized.
Choose from the following list:
cs1
cs2

cluster1::*> system cluster-switch log setup-password

Enter the switch name: cs1
RSA key fingerprint is e5:8b:c6:dc:e2:18:18:09:36:63:d9:63:dd:03:d9:cc
Do you want to continue? {y|n}::[n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> system cluster-switch log setup-password

Enter the switch name: cs2
RSA key fingerprint is 57:49:86:a1:b9:80:6a:61:9a:86:8e:3c:e3:b7:1f:b1
Do you want to continue? {y|n}:: [n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>
```

接著：`system cluster-switch log enable-collection`

```
cluster1::*> system cluster-switch log enable-collection

Do you want to enable cluster log collection for all nodes in the
cluster?
{y|n}: [n] y

Enabling cluster switch log collection.

cluster1::*>
```

如果這些命令中有任何一個出現錯誤、請聯絡NetApp支援部門。

乙太網路交換器健全狀況監控記錄收集

乙太網路交換器健全狀況監視器（CSHM）負責確保叢集與儲存網路交換器的作業健全狀況、並收集交換器記錄以供偵錯之用。本程序將引導您完成設定及開始從交換器收集詳細 * 支援 * 記錄的程序、並開始每小時收集 AutoSupport 所收集的 * 定期 * 資料。

步驟

1. 若要設定記錄收集、請針對每個交換器執行下列命令。系統會提示您輸入用於記錄收集的交換器名稱、使用者名稱和密碼。

   「系統交換器乙太網路記錄設定密碼」

   顯示範例

```
cluster1::*> system switch ethernet log setup-password
Enter the switch name: <return>
The switch name entered is not recognized.
Choose from the following list:
cs1
cs2

cluster1::*> system switch ethernet log setup-password

Enter the switch name: cs1
Would you like to specify a user other than admin for log
collection? {y|n}: n

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> system switch ethernet log setup-password

Enter the switch name: cs2
Would you like to specify a user other than admin for log
collection? {y|n}: n

Enter the password: <enter switch password>
Enter the password again: <enter switch password>
```

2. 若要啟動記錄收集、請執行下列命令、以先前命令中使用的切換參數取代裝置。這會同時啟動記錄收集的兩種類型：詳細的 * 支援 * 記錄檔和 * 定期 * 資料的每小時集合。

```
system switch ethernet log modify -device <switch-name> -log-request true
```

顯示範例

```
cluster1::*> system switch ethernet log modify -device cs1 -log
-request true

Do you want to modify the cluster switch log collection
configuration? {y|n}: [n] y

Enabling cluster switch log collection.

cluster1::*> system switch ethernet log modify -device cs2 -log
-request true

Do you want to modify the cluster switch log collection
configuration? {y|n}: [n] y

Enabling cluster switch log collection.
```

等待 10 分鐘、然後檢查記錄收集是否完成：

```
system switch ethernet log show
```

> 如果這些命令中有任何一個傳回錯誤、或記錄集合未完成、請聯絡 NetApp 支援部門。

疑難排解

如果您遇到記錄收集功能報告的下列任何錯誤狀態（可在的輸出中看到 `system switch ethernet log show`）、請嘗試對應的除錯步驟：

* 記錄收集錯誤狀態 *	* 解決方法 *
• 不存在 RSA 金鑰 *	重新產生 ONTAP SSH 金鑰。請聯絡 NetApp 支援部門。
• 交換器密碼錯誤 *	驗證認證、測試 SSH 連線、並重新產生 ONTAP SSH 金鑰。請參閱交換器說明文件、或聯絡 NetApp 支援部門以取得相關指示。
• FIPS 不存在 ECDSA 金鑰 *	如果啟用 FIPS 模式、則必須先在交換器上產生 ECDSA 金鑰、然後再重新嘗試。
• 找到之前存在的記錄 *	移除交換器上先前的記錄集合檔案。

• 交換器傾印記錄錯誤 *	確保交換器使用者擁有記錄收集權限。請參閱上述先決條件。

設定 SNMPv3

請遵循此程序來設定支援乙太網路交換器健全狀況監控（CSHM）的 SNMPv3 。

關於這項工作

下列命令可在 Cisco 92300YC 交換器上設定 SNMPv3 使用者名稱：

- 若為 * 無驗證 * ： `snmp-server user SNMPv3_USER NoAuth`
- 對於 *MD5/SHA 驗證 * ： `snmp-server user SNMPv3_USER auth [md5|sha] AUTH-PASSWORD`
- 對於採用 AES/DES 加密的 * MD5/SHA 驗證 * ： `snmp-server user SNMPv3_USER AuthEncrypt auth [md5|sha] AUTH-PASSWORD priv aes-128 PRIV-PASSWORD`

下列命令可在ONTAP Sfeside上設定一個v3使用者名稱：「cluster1：：*>安全登入create -user-or group name *MPv2_user*-applicationSNMP -imize-method USM -reme-switch-ipaddress_address_」

下列命令會使用 CSHM 建立 SNMPv3 使用者名稱： `cluster1::*> system switch ethernet modify -device DEVICE -snmp-version SNMPv3 -community-or-username SNMPv3_USER`

步驟

1. 設定交換器上的v3使用者使用驗證和加密：

   ```
   show snmp user
   ```

顯示範例

```
(sw1)(Config)# snmp-server user SNMPv3User auth md5 <auth_password>
priv aes-128 <priv_password>

(sw1)(Config)# show snmp user

-----------------------------------------------------------------
---------
                              SNMP USERS
-----------------------------------------------------------------
---------

User              Auth            Priv(enforce)   Groups
acl_filter
----------------- --------------- --------------- ---------------
-----------
admin             md5             des(no)         network-admin
SNMPv3User        md5             aes-128(no)     network-operator

-----------------------------------------------------------------
---------
     NOTIFICATION TARGET USERS (configured  for sending V3 Inform)
-----------------------------------------------------------------
---------

User              Auth                Priv
----------------- ------------------ ------------

(sw1)(Config)#
```

2. 設定位在邊上的v3使用者ONTAP ：

```
security login create -user-or-group-name <username> -application snmp
-authentication-method usm -remote-switch-ipaddress 10.231.80.212
```

顯示範例

```
cluster1::*> system switch ethernet modify -device "sw1
(b8:59:9f:09:7c:22)" -is-monitoring-enabled-admin true

cluster1::*> security login create -user-or-group-name <username>
-application snmp -authentication-method usm -remote-switch
-ipaddress 10.231.80.212

Enter the authoritative entity's EngineID [remote EngineID]:

Which authentication protocol do you want to choose (none, md5, sha,
sha2-256)
[none]: md5

Enter the authentication protocol password (minimum 8 characters
long):

Enter the authentication protocol password again:

Which privacy protocol do you want to choose (none, des, aes128)
[none]: aes128

Enter privacy protocol password (minimum 8 characters long):
Enter privacy protocol password again:
```

3. 設定 CSHM 以監控新的 SNMPv3 使用者：

```
system switch ethernet show-all -device "sw1" -instance
```

顯示範例

```
cluster1::*> system switch ethernet show-all -device "sw1" -instance

                                   Device Name: sw1
                                    IP Address: 10.231.80.212
                                  SNMP Version: SNMPv2c
                                 Is Discovered: true
   SNMPv2c Community String or SNMPv3 Username: cshm1!
                                  Model Number: N9K-C92300YC
                                Switch Network: cluster-network
                              Software Version: Cisco Nexus
Operating System (NX-OS) Software, Version 9.3(7)
                      Reason For Not Monitoring: None  <---- displays
when SNMP settings are valid
                       Source Of Switch Version: CDP/ISDP
                                 Is Monitored ?: true
                   Serial Number of the Device: QTFCU3826001C
                                   RCF Version: v1.8X2 for
Cluster/HA/RDMA

cluster1::*>
cluster1::*> system switch ethernet modify -device "sw1" -snmp
-version SNMPv3 -community-or-username <username>
cluster1::*>
```

4. 驗證要與新建立的 SNMPv3 使用者查詢的序號、是否與 CSHM 輪詢期間結束後上一步所述相同。

   ```
   system switch ethernet polling-interval show
   ```

顯示範例

```
cluster1::*> system switch ethernet polling-interval show
         Polling Interval (in minutes): 5

cluster1::*> system switch ethernet show-all -device "sw1" -instance

                                  Device Name: sw1
                                   IP Address: 10.231.80.212
                                 SNMP Version: SNMPv3
                                Is Discovered: true
   SNMPv2c Community String or SNMPv3 Username: SNMPv3User
                                 Model Number: N9K-C92300YC
                               Switch Network: cluster-network
                             Software Version: Cisco Nexus
Operating System (NX-OS) Software, Version 9.3(7)
                     Reason For Not Monitoring: None  <---- displays
when SNMP settings are valid
                      Source Of Switch Version: CDP/ISDP
                               Is Monitored ?: true
                   Serial Number of the Device: QTFCU3826001C
                                   RCF Version: v1.8X2 for
Cluster/HA/RDMA

cluster1::*>
```

版權資訊

Copyright © 2024 NetApp, Inc. 版權所有。台灣印製。非經版權所有人事先書面同意，不得將本受版權保護文件的任何部分以任何形式或任何方法（圖形、電子或機械）重製，包括影印、錄影、錄音或儲存至電子檢索系統中。

由 NetApp 版權資料衍伸之軟體必須遵守下列授權和免責聲明：

此軟體以 NETAPP「原樣」提供，不含任何明示或暗示的擔保，包括但不限於有關適售性或特定目的適用性之擔保，特此聲明。於任何情況下，就任何已造成或基於任何理論上責任之直接性、間接性、附隨性、特殊性、懲罰性或衍生性損害（包括但不限於替代商品或服務之採購；使用、資料或利潤上的損失；或企業營運中斷），無論是在使用此軟體時以任何方式所產生的契約、嚴格責任或侵權行為（包括疏忽或其他）等方面，NetApp 概不負責，即使已被告知有前述損害存在之可能性亦然。

NetApp 保留隨時變更本文所述之任何產品的權利，恕不另行通知。NetApp 不承擔因使用本文所述之產品而產生的責任或義務，除非明確經過 NetApp 書面同意。使用或購買此產品並不會在依據任何專利權、商標權或任何其他 NetApp 智慧財產權的情況下轉讓授權。

本手冊所述之產品受到一項（含）以上的美國專利、國外專利或申請中專利所保障。

有限權利說明：政府機關的使用、複製或公開揭露須受 DFARS 252.227-7013（2014 年 2 月）和 FAR 52.227-19（2007 年 12 月）中的「技術資料權利 - 非商業項目」條款 (b)(3) 小段所述之限制。

此處所含屬於商業產品和 / 或商業服務（如 FAR 2.101 所定義）的資料均為 NetApp, Inc. 所有。根據本協議提供的所有 NetApp 技術資料和電腦軟體皆屬於商業性質，並且完全由私人出資開發。 美國政府對於該資料具有非專屬、非轉讓、非轉授權、全球性、有限且不可撤銷的使用權限，僅限於美國政府為傳輸此資料所訂合約所允許之範圍，並基於履行該合約之目的方可使用。除非本文另有規定，否則未經 NetApp Inc. 事前書面許可，不得逕行使用、揭露、重製、修改、履行或展示該資料。美國政府授予國防部之許可權利，僅適用於 DFARS 條款 252.227-7015(b)（2014 年 2 月）所述權利。

商標資訊

NETAPP、NETAPP 標誌及 http://www.netapp.com/TM 所列之標章均為 NetApp, Inc. 的商標。文中所涉及的所有其他公司或產品名稱，均為其各自所有者的商標，不得侵犯。